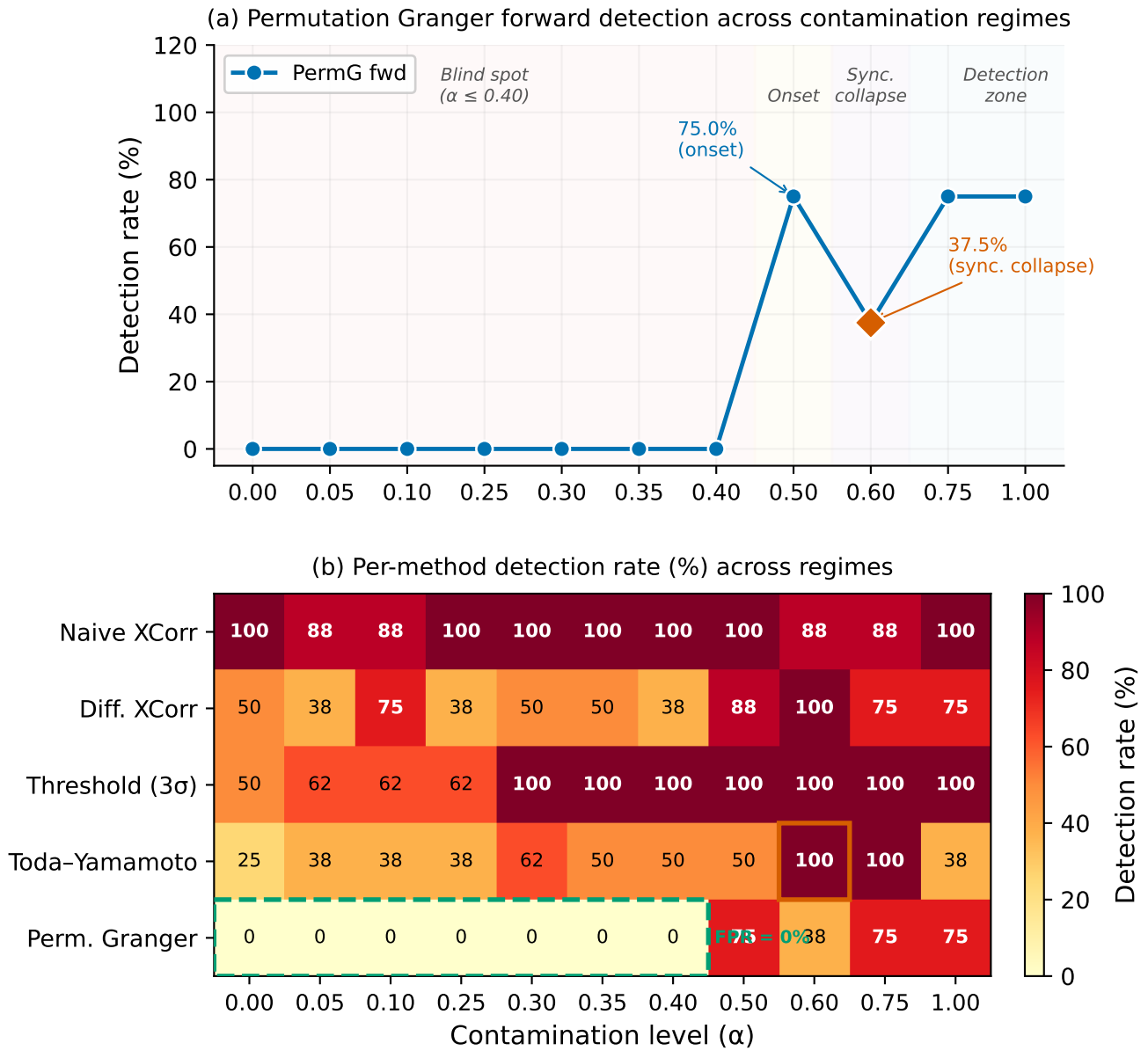
(a) Permutation Granger forward detection across contamination regimes
PermG fwd
Blind spot
(α ≤ 0.40)
Onset
Sync.
collapse
Detection
zone
75.0%
(onset)
37.5%
(sync. collapse)
Detection rate (%)
120
100
80
60
40
20
0
0.00
0.05
0.10
0.25
0.30
0.35
0.40
0.50
0.60
0.75
1.00
(b) Per-method detection rate (%) across regimes
Naive XCorr
Diff. XCorr
Threshold (3σ)
Toda–Yamamoto
Perm. Granger
100 88 88 100 100 100 100 100 88 88 100
50 38 75 38 50 50 38 88 100 75 75
50 62 62 62 100 100 100 100 100 100 100
25 38 38 38 62 50 50 50 100 100 38
0 0 0 0 0 0 0 FPR = 0% 38 75 75
Contamination level (α)
Detection rate (%)
100
80
60
40
20
0